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Device and method for controlling glide vehicle and flying body

Abstract

A glide control device includes a communications device configured to communicate with an attitude control device of a glide vehicle; and a processor configured to control the attitude control device. The processor is configured to control the attitude control device to generate a downward lift force when a velocity of the glide vehicle is higher than or equal to a first cosmic velocity.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9273943	12/2015	Poulsen	N/A	N/A
9944410	12/2017	Budd	N/A	B64D 5/00
2007/0187547	12/2006	Kelly	244/7R	B64B 1/20
2012/0223191	12/2011	Roberts	244/7A	B64C 27/24
2013/0043352	12/2012	Bahn	244/171.1	B64G 1/52
2016/0244189	12/2015	Turner	N/A	B64G 1/1085
2017/0313444	12/2016	Keidar	N/A	B64G 1/44
2018/0339793	12/2017	Chaudhary	N/A	B64G 1/625
2021/0086921	12/2020	Zhou	N/A	B64G 1/244

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110334439	12/2018	CN	N/A
WO-2014021742	12/2013	WO	B64C 30/00

OTHER PUBLICATIONS

“Falcon HTV-2”, Defense Advanced Research Projects Agency (<https://www.darpa.mil/about-us/timeline/falcon-htv-2>). cited by applicant

“Falcon Hypersonic Technology Vehicle—HTV-2”, Executive Services Directorate (https://www.esd.whs.mil/Portals/54/Documents/FOID/Reading%20Room/DARPA/14-F-0122_DARPA_X-51_DOCUMENTS.pdf). cited by applicant

“Research and Development Vision, Realization of Multi-dimension Integrated Defense Force and its Future, Supporting Documentation, Measures for Stand-off Defense Ability”, Aug. 30, 2019, Acquisition, Technology and Logistics Agency of Japan, (https://www.mod.go.jp/atla/soubiseisaku/vision/rd_vision_kaisetsu05.pdf (concise explanation of relevance provided in specification). cited by applicant

Office Action issued Sep. 26, 2023 in Japanese Patent Application No. 2020-058738, with English translation. cited by applicant

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Background/Summary

CROSS REFERENCE

(1) This application claims priority of Japanese Patent Application No. 2020-058738, filed on Mar. 27, 2020, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to glide control devices, glide vehicles, and flying bodies.

(3) Recently, flying bodies for high velocity movement between two locations on the earth have been researched. For example, a hypersonic glide vehicle, Falcon HTV2, is disclosed in “Falcon HTV-2”, Defense Advanced Research Projects Agency, and “Falcon Hypersonic Technology Vehicle—HTV-2”, Executive Services Directorate. For operating Falcon HTV-2, a rocket loaded with a hypersonic glide vehicle is launched up to an altitude of 100 km or more, where the rocket releases the hypersonic glide vehicle. The released hypersonic glide vehicle re-enters the atmosphere and glides at hypersonic velocities, for example, Mach 22 to reach a desired destination.

(4) The necessity of researches on gliding missiles configured to glide at supersonic velocities is taught in “Research and Development Vision, Realization of Multi-dimension Integrated Defense Force and its Future, Supporting Documentation, Measures for Stand-off Defense Ability”, Aug. 30, 2019, Acquisition, Technology and Logistics Agency of Japan.

BACKGROUND ART

(5) In view of the above-described situations, one of objectives of this disclosure is to provide a glide vehicle configured to reach a target in a short time. Other objectives would be understandable from the following description and explanations of embodiments.

SUMMARY

(6) In one embodiment, a glide control device includes a communications device configured to communicate with an attitude control device of a glide vehicle and a processor configured to control the attitude control device. The processor is configured to control the attitude control device to generate a downward lift force when a velocity of the glide vehicle is higher than or equal to a first cosmic velocity.

(7) In one embodiment, a glide vehicle includes the above-described glide control device and the above-described attitude control device.

(8) In one embodiment, a flying body includes the above-described glide vehicle, a propulsion device configured to accelerate the glide vehicle, and a flight control device. The flight control device is configured to control the propulsion device to achieve a level flight at a release altitude. The flight control device is further configured to release the glide vehicle at the release altitude.

(9) In one embodiment, a non-transitory tangible storage medium stores a glide program which when executed causes a processor to execute a reference altitude calculation module and an attack angle control module. The reference altitude calculation module causes the processor to calculate, based on a velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide. The attack angle control module causes the processor to generate a glide control signal to control an attitude of the glide vehicle based on the reference altitude. The attack angle control module also causes the processor to generate the glide control signal to generate a downward lift force when the velocity of the glide vehicle is higher or equal to a first cosmic velocity.

(10) The above-described embodiments allow the glide vehicle to reach a target in a short time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating an overview of a movement of a glide vehicle towards a target, according to one embodiment.
- (2) FIG. 2 is a diagram illustrating an overview of a flight of a glide vehicle at a velocity higher than or equal to the first cosmic velocity, according to one embodiment.
- (3) FIG. 3 is a diagram illustrating an overview of a flight of a glide vehicle at a velocity lower than the first cosmic velocity, according to one embodiment.
- (4) FIG. 4 is a schematic configuration diagram of a flying body, according to one embodiment.
- (5) FIG. 5 is a configuration diagram of a device configured to control a flying body, according to one embodiment.
- (6) FIG. 6 is a configuration diagram of a flight control device, according to one embodiment.
- (7) FIG. 7 is a configuration diagram of a glide vehicle, according to one embodiment.
- (8) FIG. 8 is a configuration diagram of a glide control device, according to one embodiment.
- (9) FIG. 9 is a configuration diagram of a glide program, according to one embodiment.
- (10) FIG. 10 is a flowchart illustrating a process performed in accordance with a flight program, according to one embodiment.
- (11) FIG. 11 is a diagram illustrating the direction in which a glide vehicle is ejected from a flying body, according to one embodiment.
- (12) FIG. 12 is a graph illustrating a relation between an altitude at which a glide vehicle is ejected and a flight-path angle, according to one embodiment.
- (13) FIG. 13 is a flowchart illustrating a process performed in accordance with a glide program, according to one embodiment.
- (14) FIG. 14 is a graph illustrating a relation between the velocity of a glide vehicle and a reference altitude, according to one embodiment.
- (15) FIG. 15 is a graph illustrating a relation between the velocity of a glide vehicle and the reference altitude, the relation depending on a correction factor, according to one embodiment.
- (16) FIG. 16 is a graph illustrating a relation between a correction factor and an altitude upper limit value, according to one embodiment.
- (17) FIG. 17 is a diagram illustrating an overview of a movement of a glide vehicle toward a target, according to one embodiment.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

- (18) In one embodiment, as illustrated in FIG. 1, a glide vehicle **200** is configured to take off from an on-ground facility **10** and fly toward a target facility **20**. The flight of the glide vehicle **200** from the on-ground facility **10** toward the target facility **20** includes a boost phase **500**, a glide phase **510**, and a terminal phase **520**.
- (19) During the boost phase **500**, the glide vehicle **200** is accelerated by a flying body **100** (e.g., a missile), up to a desired velocity, for example, a velocity higher than or equal to the first cosmic velocity. The flying body **100** establishes a level flight in the atmosphere at a desired altitude at which aerodynamic steering is achievable, for example, at an altitude of 60 km. The glide vehicle **200** is released from a booster **110** after the flying body **100** establishes the level flight, and the procedure goes to the glide phase **510**. The glide vehicle **200** may be released in the mesosphere (the region of altitudes from about 80 km to 50 km). It is noted that the term “level flight” referred herein may include a flight in which the angle between the flying direction and the horizontal direction is less than or equal to 10 degrees.
- (20) During the glide phase **510**, the glide vehicle **200** glides in the atmosphere, for example, in the mesosphere, with aerodynamic steering. The velocity of the glide vehicle **200** may be higher than or equal to the first cosmic velocity at the instance of time when the glide vehicle **200** is released

from the booster **110**. The glide vehicle **200** goes to the terminal phase **520** when the distance from the target facility **20** becomes less than a desired distance.

(21) During the terminal phase **520**, the glide vehicle **200** lowers the altitude and flies toward the target facility **20**.

(22) As thus described, the glide vehicle **200** is configured to move from the on-ground facility **10** to the target facility **20** such that the velocity of the glide vehicle **200** is higher than or equal to the first cosmic velocity on at least part of the route.

(23) As illustrated in FIG. 2, the glide vehicle **200** is configured to produce a downward lift force **570**, so-called negative lift force, when the glide vehicle **200** flies at a velocity higher than or equal to the first cosmic velocity. While flying at a velocity higher than or equal to the first cosmic velocity, the glide vehicle **200** receives a centrifugal force **550** larger than the gravity **560** (the force exerted on the glide vehicle **200** by the earth due to the law of universal gravitation). The inventors have found that the glide vehicle **200** can fly in the horizontal direction **610** by producing a lift force **570** in a downward direction (e.g., the gravity direction). Accordingly, in embodiments where the glide vehicle **200** is configured to generate a lift force by controlling the attitude of the glide vehicle **200** or the attitude of part of the glide vehicle **200**, a negative lift force is produced by controlling the attitude while the glide vehicle **200** is flying at velocities higher than or equal to the first cosmic velocity. To produce a negative lift force by controlling the attitude of the glide vehicle **200**, for example, the forward direction **600** (e.g., the wind chord direction) of the glide vehicle **200** is tilted downward with respect to the flying direction of the glide vehicle **200**, e.g., the horizontal direction **610**. The glide vehicle **200** is controlled such that the angle of attack (AoA) **602** is tilted downward. The angle of attack **620** is defined as positive when the angle of attack **620** is tilted upward, and as negative when the angle of attack **620** is tilted downward. The angle of attack **620** may be set to a negative value while the glide vehicle **200** is flying at a velocity higher than or equal to the first cosmic velocity.

(24) The velocity of the glide vehicle **200** may decrease due to the drag force during the glide phase **510**.

(25) When the glide vehicle **200** flies at a velocity lower than the first cosmic velocity, the glide vehicle **200** flies such that an upward lift force **570** is produced as illustrated in FIG. 3. While flying at a velocity lower than the first cosmic velocity, the centrifugal force **550** received by the glide vehicle **200** is smaller than the gravity **560**. The glide vehicle **200** can fly in the horizontal direction **610** by producing the lift force **570** upward. Accordingly, the forward direction **600** of the glide vehicle **200** is tilted upward with respect to the flying direction of the glide vehicle **200**, for example, the horizontal direction **610**, when the glide vehicle **200** flies at a velocity lower than the first cosmic velocity. The glide vehicle **200** is controlled such that the angle of attack **620** is tilted upward.

(26) Flying Body Configuration

(27) The glide vehicle **200** is loaded in the flying body **100**. As illustrated in FIG. 4, the flying body **100** includes a booster **110**, a nose faring **120**, and the glide vehicle **200**. The booster **110** is configured to accelerate the flying body **100** and control the attitude of the flying body **100**. For example, the booster **110** may include one or more rocket engines. The booster **110** is configured to accelerate the flying body **100** up to a desired velocity, for example, a velocity higher than or equal to the first cosmic velocity during the boost phase **500**.

(28) The nose faring **120** is disposed at the tip of the flying body **100** to cover the glide vehicle **200**. The nose faring **120** is configured to protect the glide vehicle **200** from raindrops and the dynamic pressure produced during the boost phase **500**.

(29) As illustrated in FIG. 5, the booster **110** includes an inertial measurement unit **111**, a flight control device **112**, a propulsion device **113**, and a release device **114**.

(30) The release device **114** is configured to detach the nose faring **120** and the glide vehicle **200** from the booster **110** upon the transition from the boost phase **500** to the glide phase **510**. The

release device **114** may include a spring, for example.

(31) The inertial measurement unit **111** is configured to measure the status of the flying body **100**, including the altitude, the flying direction, and the velocity. For example, the inertial measurement unit **111** may include an acceleration sensor, an angular acceleration sensor, and/or the like to measure inertial information of the flying body **100**, including the acceleration and the angular acceleration. The status of the flying body **100** (e.g., the altitude, the flying direction, and the velocity) is determined based on the measured inertial information. A signal indicative of the determined status of the flying body **100** is sent to the flight control device **112**. The inertial measurement unit **111** may include a receiver receiving a global positioning system (GPS) signal to determine the status of the flying body **100** based on the GPS signal.

(32) The flight control device **112** is configured to control the propulsion device **113** and the release device **114** based on the status of the flying body **100** acquired from the inertial measurement unit **111**. The flight control device **112** is electrically connected to the inertial measurement unit **111**, the propulsion device **113**, and the release device **114** with wire or wirelessly and configured to receive the signal indicative of the status of the flying body **100** from the inertial measurement unit **111** and send control signals to the propulsion device **113** and the release device **114**. As illustrated in FIG. 6, the flight control device **112** includes a communications device **151**, an input/output device **152**, a storage device **153**, and a processor **154**.

(33) The communications device **151** is electrically connected to the inertial measurement unit **111**, the propulsion device **113**, and the release device **114** and configured to communicate with these devices. The communications device **151** is configured to forward data received from the inertial measurement unit **111** to the processor **154**. The communications device **151** is further configured to forward signals generated by the processor **154** to the propulsion device **113** and the release device **114**. The communications device **151** may include various interfaces, such as a network interface card (NIC) and a universal serial bus (USB).

(34) The input/output device **152** is configured to receive information used for processing performed by the processor **154**. The input/output device **152** is further configured to output the result of the processing performed by the processor **154**. The input/output device **152** may include various input devices and output devices, for example, a keyboard, a mouse, a microphone, a display, a speaker, a touch panel, and the like. The input/output device **152** may be omitted.

(35) The storage device **153** is configured to store various data for controlling the propulsion device **113** and the release device **114**, including a flight program **300**. The storage device **153** is used as a non-transitory tangible storage medium that stores the flight program **300**. The flight program **300** may be provided in the form of a computer program product recorded on a computer-readable storage medium **30** or in the form of a computer program product downloadable from a server.

(36) The processor **154** is configured to perform various data processes to control the propulsion device **113** and the release device **114**. The processor **154** is configured to read out the flight program **300** from the storage device **153** and execute the flight program **300** to generate control signals for controlling the propulsion device **113** and the release device **114**. The processor **154** may include, for example, a central processing unit (CPU) or the like.

(37) Glide Vehicle Configuration

(38) The glide vehicle **200** includes a heat-resistant structure tolerant of aerodynamic heating caused by a flight at the first cosmic velocity in the atmosphere, for example, in the mesosphere. As illustrated in FIG. 7, the glide vehicle **200** includes an inertial measurement unit **210**, a glide control device **220**, and an attitude control device **230**.

(39) The inertial measurement unit **210** is configured to measure the location, velocity, and altitude of the glide vehicle **200**. The inertial measurement unit **210** is configured to send a signal indicative of the velocity and altitude of the glide vehicle **200** to the glide control device **220**. The inertial measurement unit **210** may include an acceleration sensor, an angular acceleration sensor, and/or

the like. The inertial measurement unit **210** may be configured to determine the velocity of the glide vehicle **200** based on the acceleration measured by the acceleration sensor and the angular acceleration measured by the angular acceleration sensor. In such embodiments, the inertial measurement unit **210** is configured to determine the velocity of the glide vehicle **200** based on the history of the measured accelerations and angular accelerations.

(40) The inertial measurement unit **210** may be further configured to determine the altitude of the glide vehicle **200** based on the acceleration measured by the acceleration sensor and the angular acceleration measured by the angular acceleration sensor. In such embodiments, the inertial measurement unit **210** is configured to determine the altitude of the glide vehicle **200** based on the history of the measured accelerations and angular accelerations.

(41) The attitude control device **230** is configured to control the attitude of the glide vehicle **200**. The attitude control device **230** may be configured to control, for example, the angle of attack **620** of the glide vehicle **200**.

(42) The glide control device **220** is configured to control the attitude control device **230** based on the altitude and velocity of the glide vehicle **200** acquired from the inertial measurement unit **210**. The glide control device **220** is electrically connected to the inertial measurement unit **210** and the attitude control device **230** with wire or wirelessly and configured to receive the signal indicative of the altitude and velocity of the glide vehicle **200** from the inertial measurement unit **210** and send a control signal to the attitude control device **230**. As illustrated in FIG. 8, the glide control device **220** includes a communications device **221**, an input/output device **222**, a storage device **223**, and a processor **224**.

(43) The communications device **221** is electrically connected to the inertial measurement unit **210** and the attitude control device **230** and configured to communicate with these devices. The communications device **221** is configured to forward data received from the inertial measurement unit **210** to the processor **224**. The communications device **221** is further configured to forward a signal generated by the processor **224** to the attitude control device **230**. The communications device **221** may include various interfaces, such as a network interface card (NIC) and a universal serial bus (USB).

(44) The input/output device **222** is configured to receive information used for processing performed by the processor **224**. The input/output device **222** is further configured to output the result of the processing performed by the processor **224**. The input/output device **222** may include various input devices and output devices, for example, a keyboard, a mouse, a microphone, a display, a speaker, a touch panel, and the like. The input/output device **222** may be omitted.

(45) The storage device **223** is configured to store various data for controlling the attitude control device **230** of the glide vehicle **200**, including a glide program **400**. The storage device **223** is used as a non-transitory tangible storage medium that stores the glide program **400**. The glide program **400** may be provided in the form of a computer program product recorded on a computer-readable storage medium **40** or in the form of a computer program product downloadable from a server. A single storage medium may be commonly used as the storage medium **30** and the storage medium **40**.

(46) The processor **224** is configured to perform various data processes to control the attitude control device **230** of the glide vehicle **200**. The processor **224** is configured to read out the glide program **400** from the storage device **223** and execute the glide program **400** to generate a control signal for controlling the attitude control device **230**. The processor **224** may include, for example, a central processing unit (CPU) or the like.

(47) The processor **224** is configured to, by executing the glide program **400**, generate an attitude control signal that controls the attitude control device **230** such that a lift force **570** is produced downward, for example, in the gravity direction, when the velocity of the glide vehicle **200** is higher than or equal to the first cosmic velocity. As illustrated in FIG. 9, the glide program **400** includes a reference altitude calculation module **410**, an attack angle control module **420**, and a

descent control module **430**.

(48) The processor **224** is configured to execute the reference altitude calculation module **410** and thereby calculate, based on the velocity **211** of the glide vehicle **200**, a reference altitude at which the glide vehicle **200** should fly. The velocity **211** of the glide vehicle **200** is acquired from the inertial measurement unit **210**.

(49) The processor **224** is configured to execute the attack angle control module **420** and thereby generate the attitude control signal for controlling the attitude control device **230** based on the altitude **212** of the glide vehicle **200** and the reference altitude calculated through the execution of the reference altitude calculation module **410**. The altitude **212** of the glide vehicle **200** is acquired from the inertial measurement unit **210**. The processor **224** is configured to determine the angle of attack **620** of the glide vehicle **200** based on the altitude **212** of the glide vehicle **200** and the determined reference altitude. The processor **224** is further configured to generate the attitude control signal based on the determined angle of attack **620**.

(50) The processor **224** is further configured to cause the glide vehicle **200** to fly toward the target facility **20** by executing the descent control module **430**. The processor **224** is configured to control the attitude control device **230** to lower the altitude of the glide vehicle **200** when the glide vehicle **200** gets close to the target facility **20**. The attitude control device **230** is configured to guide the glide vehicle **200** to the target facility **20** under the control of the processor **224**.

(51) Operation of Flight Control Device

(52) The processor **154** of the flight control device **112** is configured to execute the flight program **300** when the flying body **100** takes off the on-ground facility **10**. The flight program **300** causes the processor **154** to perform the procedure illustrated in FIG. **10**. The processor **154** is configured to control the flying body **100** during the boost phase **500** by using the flight program **300**.

(53) At step **S110**, the processor **154** controls the propulsion device **113** to cause the flying body **100** to take off and accelerate the velocity. The propulsion device **113** is controlled such that the flying body **100** achieves a level flight at an altitude at which the glide vehicle **200** is released (hereinafter, referred to as “release altitude”.) The release altitude may be 30 km or higher, for example. The release altitude may be 50 km or higher. In other embodiments, the release altitude may be 60 km or higher. The release altitude may be 100 km or lower. In other embodiments, the release altitude may be 80 km or lower.

(54) At step **S120**, the processor **154** determines whether the flying body **100** reaches the release altitude. The altitude of the flying body **100** is determined by the inertial measurement unit **111**. The processor **154** acquires the altitude of the flying body **100** from the inertial measurement unit **111**. When the acquired altitude of the flying body **100** is lower than the release altitude (NO at step **120**), the processor **154** waits for the flying body **100** to reach the release altitude. When the acquired altitude of the flying body **100** becomes higher than or equal to the release altitude (YES at step **S120**), the processor **154** proceeds with the process of step **S130**.

(55) At step **S130**, the processor **154** generates a nose faring detachment signal to detach the nose faring **120**. The release device **114** detaches the nose faring **120** based on the nose faring detachment signal.

(56) At step **S140**, the processor **154** controls the release device **114** to eject the glide vehicle **200**. The processor **154** generates an ejection signal that requests an ejection of the glide vehicle **200**. The release device **114** ejects the glide vehicle **200** based on the ejection signal. For example, when releasing the glide vehicle **200**, the flying body **100** ejects the glide vehicle **200** while flying in an ejection direction **650** tilted downward by a flight-path angle **660** with respect to the horizontal direction **610**, as illustrated in FIG. **11**. The angle of attack **620** of the glide vehicle **200** may be arbitrarily selected immediately after the ejection. The angle of attack **620** may be 0 degrees, for example.

(57) As illustrated in FIG. **12**, the ejection direction **650** in which the glide vehicle **200** is ejected is determined depending on the release altitude. The flight-path angle **660**, which is the angle between

the ejection direction **650** and the horizontal direction **610**, falls in a flight-path angle range **830** between a minimum flight-path angle **810** and a maximum flight-path angle **820**. The minimum flight-path angle **810** is the minimum value of the flight-path angle **660** that allows the glide vehicle **200** to glide in the atmosphere. When the flight-path angle **660** is smaller than the minimum flight-path angle **810**, the glide vehicle **200** may get rid of the atmosphere. The maximum flight-path angle **820** is the maximum value of the flight-path angle **660** that allows the glide vehicle **200** to fly over a desired distance. When the flight-path angle **660** is larger than the maximum flight-path angle **820**, the glide vehicle **200** may be decelerated, which may make it impossible to fly over the desired distance.

(58) The minimum flight-path angle **810** monotonically increases as the altitude increases. The function that represents the correlation of the minimum flight-path angle **810** with the altitude may include a convex upward function defined such that the increase rate of the minimum flight-path angle **810** decreases as the altitude increases.

(59) The maximum flight-path angle **820** also monotonically increases as the altitude increases as is the case with the minimum flight-path angle **810**. The function that represents the correlation of the maximum flight-path angle **820** with the altitude may include a convex upward function defined such that the increase rate of the maximum flight-path angle **820** decreases as the altitude increases.

(60) The release altitude and the ejection direction **650** may be determined based on the velocity of the flying body **100** at the instance of time when the glide vehicle **200** is released. The flight-path angle range **830** may be determined based on the velocity of the flying body **100** at the instance of time when the glide vehicle **200** is released. For example, for a fixed release altitude, the lower limit and/or the upper limit of the flight-path angle range **830** may increase as the velocity of the glide vehicle **200** increases. When the velocity of the glide vehicle **200** at the instance of time when the glide vehicle **200** is 15 km per second, as illustrated in FIG. 12, the flight-path angle range **830** is, for example, larger than or equal to -7 degrees. The flight-path angle range **830** may be smaller than or equal to 10 degrees. For release altitudes of 40 km or higher, the flight-path angle range **830** may be larger than or equal to -3 degrees. For release altitudes of 50 km or higher, the flight-path angle range **830** may be larger than or equal to 1 degree. For release altitudes of 60 km or higher, the flight-path angle range **830** may be larger than or equal to 3 degrees. For release altitudes of 80 km or lower, the flight-path angle range **830** may be smaller than or equal to 9 degrees. The combination of the release altitude and the ejection direction **650** may be selected such that the flight-path angle falls in the flight-path angle range **830**.

(61) As discussed above, the processor **154** of the flight control device **112** allows releasing the glide vehicle **200** in the atmosphere by executing the flight program **300**. The velocity of the flying body **100** may be higher than or equal to the first cosmic velocity at the instance of time when the glide vehicle **200** is released. The glide vehicle **200** can reach the target facility **20** in a short time since the glide vehicle **200** can be released in the atmosphere when the flying body **100** is flying at a velocity higher than or equal to the first cosmic velocity.

(62) Operation of Glide Control Device

(63) The processor **224** of the glide control device **220** is configured to execute the glide program **400** when the glide vehicle **200** is released from the booster **110**. The glide program **400** is configured to cause the processor **224** to perform the procedure illustrated in FIG. 13. The processor **224** is configured to control the glide vehicle **200** during the glide phase **510** and the terminal phase **520** in accordance with the glide program **400**.

(64) At step **S210**, the processor **224** acquires the current velocity and altitude of the glide vehicle **200**. The current velocity and altitude of the glide vehicle **200** are measured by the inertial measurement unit **210**. The processor **224** acquires the current velocity and altitude of the glide vehicle **200** from the inertial measurement unit **210**.

(65) At step **S220**, the processor **224** determines, based on the current velocity of the glide vehicle **200**, a reference altitude at which the glide vehicle **200** should glide. As illustrated in FIG. 12, the

reference altitude may be represented as a function of the current velocity of the glide vehicle **200**. For example, the reference altitude may be determined such that the gravity **560**, the lift force **570**, and the centrifugal force **550** exerted on the glide vehicle **200** are balanced in the vertical direction (e.g., the gravity direction). The reference altitude may be determined such that the lift-drag ratio of the glide vehicle **200** becomes maximum. Details of the reference altitude will be described later.

(66) At step **S230**, the processor **224** controls the attitude control device **230** such that the glide vehicle **200** glides at the reference altitude. The processor **224** compares the current altitude of the glide vehicle **200** acquired from the inertial measurement unit **210** with the reference altitude. Based on the comparison between the current altitude and the reference altitude, the processor **224** generates an attitude control signal to control the attitude control device **230**. When the current altitude is higher than the reference altitude, for example, the processor **224** controls the attitude control device **230** to decrease the angle of attack **620**. When the current altitude is lower than the reference altitude, the processor **224** controls the attitude control device **230** to increase the angle of attack **620**. Accordingly, when the glide vehicle **200** glides at a velocity higher than the first cosmic velocity, the processor **224** controls the attitude control device **230** to decrease the angle of attack **620** such that a negative lift force is produced, since the altitude of the glide vehicle **200** will increase as the angle of attack **620** increases. The processor **224** may control the attitude control device **230** based on a proportional-differential (PD) control using the current altitude and the reference altitude. The processor **224** may control the attitude control device **230** based on the angle of attack **620** that achieves the maximum lift-drag ratio, when the calculated angle of attack **620** is larger than the angle of attack **620** that achieves the maximum lift-drag ratio. For example, when the calculated angle of attack **620** is larger than the angle of attack **620** that achieves the maximum lift-drag ratio, the processor **224** may control the attitude control device **230** such that the angle of attack **620** of the glide vehicle **200** is adjusted to achieve the maximum lift-drag ratio.

(67) At step **S240**, the processor **224** determines whether the glide vehicle **200** reaches a target location at which the glide vehicle **200** starts to descend toward the target facility **20**. The processor **224** compares the location of the glide vehicle **200** with the target location. The processor **224** may determine that the glide vehicle **200** reaches the target location, when the distance between the location of the glide vehicle **200** and the target location is less than a desired value. The target location may be defined as a region in which the distance from the target facility **20** is less than a desired distance. In such embodiments, the processor **224** may determine that the glide vehicle **200** reaches the target location, when the location of the glide vehicle **200** falls in the region corresponding to the target location. When the glide vehicle **200** does not reach the target location (NO at step **S240**), the processor **224** performs the process of step **S210** and repeats steps **S210** to **S240** until the glide vehicle **200** reaches the target location. When the glide vehicle **200** reaches the target location (YES at step **S240**), the processor **224** performs the process of step **S250**.

(68) The processor **224** acquires the location of the glide vehicle **200** from the inertial measurement unit **210**. The inertial measurement unit **210** may include, for example, an acceleration sensor, an angular acceleration sensor, and the like and may be configured to determine the location of the glide vehicle **200** based on the history of the accelerations measured by the acceleration sensor and the history of the angular accelerations measured by the angular acceleration sensor.

(69) At step **S250**, to go to the terminal phase **520**, the processor **224** controls the attitude control device **230** to reduce the altitude of the glide vehicle **200** toward the target facility **20**. For example, the processor **224** may generate the attitude control signal to decrease the angle of attack **620** such that a negative lift force is exerted on the glide vehicle **200**. The glide vehicle **200** starts to descend as the attitude control device **230** decreases the angle of attack **620** based on the attitude control signal. The processor **224** controls the attitude control device **230** such that the glide vehicle **200** reaches the target facility **20**. The method to cause the glide vehicle **200** to descend toward the target facility **20** may be arbitrarily selected.

(70) As thus described, the processor **224** of the glide control device **220** allows the glide vehicle

200 to fly in the atmosphere even when the velocity of the glide vehicle **200** is higher than or equal to the first cosmic velocity. Accordingly, the glide vehicle **200** can reach the target facility **20** in a short time. The glide vehicle **200** may lack a device to produce propulsion to accelerate in the flying direction during the glide phase **510**. It is noted that the reference altitude calculation module **410** causes the processor **224** to perform the processes of steps **S210** and **S220**. The attack angle control module **420** causes the processor **224** to perform the process of step **S230**. The descent control module **430** causes the processor **224** to perform the processes of steps **S240** and **S250**.

(71) Reference Altitude

(72) As illustrated in FIG. **14**, for velocities of the glide vehicle **200** lower than the first cosmic velocity **730**, the function representing the correlation of the reference altitude with the velocity of the glide vehicle **200** may be a weakly monotonically increasing function. For velocities of the glide vehicle **200** higher than the first cosmic velocity **730**, the function representing the correlation of the reference altitude with the velocity of the glide vehicle **200** may be a weakly monotonically decreasing function. For velocities of the glide vehicle **200** higher than the first cosmic velocity **730**, the function representing the correlation of the reference altitude with the velocity of the glide vehicle **200** may be a convex downward function for which the decrease rate decreases as the velocity of the glide vehicle **200** increases.

(73) An altitude upper limit value **700** may be determined for the reference altitude. The reference altitude may be determined as the altitude upper limit value **700** when the calculated reference altitude is higher than the altitude upper limit value **700**. The altitude upper limit value **700** may be an altitude that falls in the mesosphere. The altitude upper limit value **700** may be higher than or equal to 50 km. The altitude upper limit value **700** may be higher than or equal to 60 km. The altitude upper limit value **700** may be lower than or equal to 80 km. The altitude upper limit value **700** may be lower than or equal to 70 km.

(74) The reference altitude may be calculated in view of the fact that the gravity **560**, the lift force **570**, and the centrifugal force **550** exerted on the glide vehicle **200** are balanced in the vertical direction. The reference altitude may be calculated, for example, in accordance with the following expression (1):

$$(75) \quad h = KH_e \ln\left(\frac{\rho_0 R_e}{m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2}\right), \quad (1)$$

where h is the reference altitude; K is the correction factor; $H_{\text{sub.e}}$ is the scale height of the earth used to calculate the density of the atmospheric air; $\rho_{\text{sub.0}}$ is the air density at an altitude of 0 m; R_e is the earth radius; $\beta_{\text{sub.m}}$ is the ballistic coefficient; $V_{\text{sub.e}}$ is the first cosmic velocity **730**; and $L/D_{\text{sub.max}}$ is the maximum lift-drag ratio of the glide vehicle **200**.

(76) A description is given below of how the expression (1) is derived. Taking account of a negative lift force, since the lift force **570** is balanced with the resultant of the gravity **560** and the centrifugal force **550**, the lift force **570** exerted on the glide vehicle **200** is approximately represented by the following expression (2):

$$(77) \quad L = |mg - \frac{mV^2}{R_e} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2}|, \quad (2)$$

where L is the lift force **570** of the glide vehicle **200**; m is the mass of the glide vehicle **200**; g is the acceleration of gravity; and V is the velocity of the glide vehicle **200**.

(78) The lift force **570** of the glide vehicle **200** is calculated in accordance with the following expression (3):

$$(79) \quad L = \frac{1}{2} \rho V^2 S C_L, \quad (3)$$

where ρ is the air density; S is the area of the member (e.g. a wing) that produces the lift force **570**; and $C_{\text{sub.L}}$ is the lift coefficient of the glide vehicle **200**.

(80) According to “Dynamics of Atmospheric Reentry”, Frank J. Regan, AIAA Education Series, the ballistic coefficient is calculated in accordance with expression (4):

$$(81) \quad m = \frac{m}{C_{D0} S}, \quad (4)$$

where C.sub.D0 is the zero lift drag coefficient.

(82) The maximum lift-drag ratio is calculated in accordance with expression (5):

$$(83) \quad L / D_{\max} = \frac{C_L^*}{C_D^*}, \quad (5)$$

where C.sub.L* is the lift coefficient for the case where the lift-drag ratio is maximum, and C.sub.D* is the drag coefficient for the case where the lift-drag ratio is maximum.

(84) According to “Dynamics of Atmospheric Reentry”, Frank J. Regan, AIAA Education Series, the drag coefficient for the case where the lift-drag ratio is maximum is approximately represented, using the zero lift drag coefficient, by expression (6):

$$(85) \quad C_D^* = 2C_{D0}. \quad (6)$$

(86) The lift force **570** for the case where the lift-drag ratio is maximum can be calculated in accordance with expression (7), which is acquired by substituting expressions (4) to (6) for expression (3):

$$(87) \quad L = m V^2 / D_{\max}^{-1}. \quad (7)$$

(88) The first cosmic velocity **730** is calculated by expression (8):

$$(89) \quad V_e = \sqrt{gR_e}. \quad (8)$$

(90) Expression (9) is acquired by substituting expression (8) for expression (2) as follows:

$$(91) \quad L = \left| \frac{mV_e^2}{R_e} - \frac{mV^2}{R_e} \right|. \text{Math.} \quad (9)$$

(92) According to “Dynamics of Atmospheric Reentry”, Frank J. Regan, AIAA Education Series, the air density is calculated by expression (10):

$$(93) \quad \rho = \rho_0 e^{-\frac{h}{H_e}}. \quad (10)$$

(94) Expression (11) is acquired by substituting expressions (9) and (10) for expression (7) as follows:

$$(95) \quad m \rho e^{-\frac{h}{H_e}} L / D_{\max}^{-1} = \left| \frac{mV_e^2}{R_e} - \frac{mV^2}{R_e} \right|. \quad (11)$$

(96) Expression (12) is acquired by rewriting expression (11) as follows:

$$(97) \quad h = H_e \ln \left(\frac{\rho_0 R_e}{m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2} \cdot \text{Math.} \right). \quad (12)$$

(98) Expression (12) represents an equilibrium altitude at which the lift force **570** and the centrifugal force **550** are balanced in the vertical direction for the current velocity. Accordingly, the reference altitude is set to an altitude lower than the equilibrium altitude determined in accordance with expression (12). Expression (1) is acquired by multiplying the right side of expression (12) by a correction factor K smaller than one, which aims at achieving a stable flight of the glide vehicle **200**. The reference altitude may be calculated by expression (13), in which a desired value α is subtracted from the right side of expression (12):

$$(99) \quad h = H_e \ln \left(\frac{\rho_0 R_e}{m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2} \cdot \text{Math.} \right) - \alpha. \quad (13)$$

In such embodiments, a lower limit value may be defined since the reference altitude may be calculated as zero or below for reduced velocities. The reference altitude may be determined as the lower limit value when the calculated reference altitude is lower than the lower limit value.

Correction Factor

(100) The correction factor K may be in a desired range. The flying distance of the glide vehicle **200** decreases as the correction factor K decreases. For example, the correction factor K may be larger than 0.7. The correction factor K may be larger than or equal to 0.8. The correction factor K may be larger than 0.8. When the correction factor K is increased, the altitude of the flying body **100** may exceed the equilibrium altitude at which the lift force **570** and the centrifugal force **550**

are balanced in the vertical direction. The correction factor K may be smaller than or equal to 0.95. The correction factor K may be smaller than 0.95. The correction factor K may be smaller than or equal to 0.9. The correction factor K may be smaller than 0.9.

(101) The correction factor K may be determined based on the altitude upper limit value **700**. As illustrated in FIG. **15**, the reference altitude increases as the correction factor K increases. Accordingly, the probability that the altitude of the flying body **100** exceeds the equilibrium altitude increases as the correction factor K increases.

(102) As illustrated in FIG. **16**, the upper limit value **760** of the correction factor K may be determined based on the altitude upper limit value **700**. The upper limit value **760** of the correction factor K may weakly monotonically decreases against the altitude upper limit value **700**. In embodiments where the altitude upper limit value **700** is larger than 70 km, for example, the upper limit value **760** of the correction factor K may decrease as the altitude upper limit value **700** increases. The altitude of the flying body **100** may exceed the equilibrium altitude when the altitude upper limit value **700** is larger than 70 km and the correction factor K is 0.9. In embodiments where the altitude upper limit value **700** is larger than 70 km, for example, the function that represents the correlation of the upper limit value **760** of the correction factor K with the altitude upper limit value **700** may be a linear function.

(103) The correction factor K may be in a range **770** in which the flying body **100** can fly stably with high efficiency. The correction factor K which falls in the range **770** may be smaller than or equal to the upper limit value **760** of the correction factor K determined based on the altitude upper limit value **700**. The correction factor K which falls in the range **770** may be larger than 0.8. The flying body **100** can fly with high efficiency when the correction factor K is small and the altitude upper limit value **700** is large. In embodiments where the altitude upper limit value **700** is in a range from 50 km to 60 km, for example, the lower limit value **780** of the correction factor K which falls in the range **770** may be determined based on the altitude upper limit value **700**. In embodiments where the altitude upper limit value **700** is in a range from 50 km to 60 km, the lower limit value **780** of the correction factor K may decrease as the altitude upper limit value **700** increases. In embodiments where the altitude upper limit value **700** is in a range from 50 km to 60 km, the function that represents the correlation of the lower limit value **780** of the correction factor K with the altitude upper limit value **700** may be a linear function.

Modification Examples

(104) The flying body **100** may be ejected from a moving object, such as an aircraft. In such embodiments, the flying body **100** ejected from the aircraft may be configured to release the glide vehicle **200** in the atmosphere, for example, in the mesosphere.

(105) In the procedure illustrated in FIG. **10**, the nose faring **120** may be detached before the flying body **100** reaches the releasing altitude. The nose faring **120**, which is required only to protect the glide vehicle **200** from the dynamic pressure and raindrops, may be detached when the flying body **100** reaches an altitude at which the effect of the atmospheric air is small.

(106) The flight control device **112** may be configured to acquire the location, velocity, and altitude of the flying body **100** from the inertial measurement unit **210** of the glide vehicle **200**. In such embodiments, the inertial measurement unit **111** of the booster **110** may be omitted.

(107) The flight program **300** may be executed by the processor **224** of the glide vehicle **200**. In such embodiments, the processor **224** of the glide vehicle **200** may be configured to control the propulsion device **113** and the release device **114**.

(108) The processor **224** of the glide vehicle **200** may be configured to acquire the altitude and velocity of the glide vehicle **200** from an entity external to the glide vehicle **200**. For example, as illustrated in FIG. **17**, a detect device **50** (e.g., a radar device) may be configured to detect the glide vehicle **200** and measure the altitude and velocity of the glide vehicle **200**. In such embodiments, the detect device **50** may be configured to send the measured altitude and velocity to the glide vehicle **200**. The inertial measurement unit **210** may be configured to repeatedly receive a GPS

signal to calculate the altitude and velocity. The velocity may be calculated based on the history of positional information included in the GPS signal.

(109) The release altitude may be an altitude out of the atmosphere, for example, an altitude higher than 100 km. When the releasing altitude is an altitude out of the atmosphere, the flying direction of the flying body **100** may be tilted downward with respect to the horizontal direction **610** at the release altitude.

(110) The above-described embodiments and modification examples are mere examples and may be modified as long as their functionalities are maintained. The configurations presented in the embodiments and the modification examples may be arbitrarily modified and/or combined as long as their functionalities are maintained.

(111) The glide control devices set forth in the above-described embodiments are represented, for example, as follows.

(112) In a first aspect, a glide control device includes a communications device and a processor. The processor is configured to control an attitude control device to generate a downward lift force when a velocity of the glide vehicle is higher than or equal to the first cosmic velocity.

(113) The glide control device thus configured allows the glide vehicle to reach a target in a short time since the glide vehicle can fly at a velocity higher than or equal to the first cosmic velocity in the atmosphere.

(114) In a second aspect, the glide control device in the first aspect is configured such that the processor determines, based on the velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide and controls the attitude control device based on a current altitude of the glide vehicle and the reference altitude.

(115) The glide control device thus configured can control the attitude of the glide vehicle based on the altitude, thereby suppressing a decrease in the flying distance of the glide vehicle caused by an unnecessary deceleration.

(116) In a third aspect, the glide control device in the second aspect is configured such that the reference altitude is determined such that a gravity, a lift force, and a centrifugal force which are exerted on the glide vehicle are balanced in a gravity direction.

(117) In a fourth aspect, the glide control device in the second aspect is configured such that the reference altitude is calculated in accordance with the above-described expression (1).

(118) In a fifth aspect, the glide control device in the second aspect is configured such that an altitude upper limit value is defined for the reference altitude.

(119) This allows the glide vehicle to stably glide at velocities near the first cosmic velocity.

(120) In a sixth aspect, a glide vehicle includes the above-described glide control device and the above-described attitude control device.

(121) In a seventh aspect, the glide vehicle in the sixth aspect lacks a propulsion device configured to accelerate the glide vehicle.

(122) In an eighth aspect, a flying body includes: the glide vehicle in the seventh aspect; a propulsion device configured to accelerate the glide vehicle; and a flight control device. The flight control device is configured to control the propulsion device to achieve a level flight at a release altitude. The flight control device is further configured to release the glide vehicle at the release altitude.

(123) In a ninth aspect, the flying body in the eighth aspect is configured such that the flight control device controls the propulsion device such that a flying direction of the flying body is tilted downward with respect to a horizontal direction at the release altitude when the release altitude is an altitude out of the atmosphere.

(124) In a tenth aspect, the flying body in the eighth aspect is configured such that the release altitude is an altitude in the atmosphere at which aerodynamic steering is achievable.

(125) The flying body thus configured allows the glide vehicle to reach a target in a short time as the glide vehicle is released in the atmosphere.

(126) In an eleventh aspect, the flying body in the eighth aspect is configured such that the release altitude includes a reference altitude at which the glide vehicle should glide.

(127) The flying body thus configured allows the glide vehicle to reach a target in a short time as the glide vehicle can reach the reference altitude without largely changing the altitude.

(128) In a twelfth aspect, the flying body in the eighth aspect is configured such that the flight control device controls the propulsion device to allow the flying body to have a velocity higher than or equal to the first cosmic velocity at the release altitude.

(129) In a thirteen aspect, a non-transitory tangible storage medium stores a program which when executed causes a processor to execute a reference altitude calculation module and an attack angle control module. The reference altitude calculation module causes the processor to calculate, based on a velocity of a glide vehicle, a reference altitude at which the glide vehicle should glide. The attack angle control module causes the processor to generate, based on the reference altitude, a glide control signal that controls an attitude of the glide vehicle. The attack angle control module causes the processor to generate the glide control signal to produce a downward lift force when the velocity is higher than or equal to a first cosmic velocity.

Claims

1. A glide control device, comprising: a communications device configured to communicate with an attitude control device of a glide vehicle in which no propulsion device configured to accelerate the glide vehicle is disposed; and a processor configured to control the attitude control device, wherein the processor is configured to control the attitude control device to generate a downward lift force so as to allow the glide vehicle to fly in a horizontal direction when a velocity of the glide vehicle is higher than or equal to a first cosmic velocity, and wherein the first cosmic velocity is a velocity required by a spacecraft in a circular low Earth orbit, and wherein the processor is configured to: determine, based on the velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide; and control the attitude control device based on a current altitude of the glide vehicle and the reference altitude, wherein the reference altitude is calculated in accordance with the following expression (1):
$$h = KH_e \ln\left(\frac{R_e}{m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2}\right), \quad (1)$$
 wherein h is the reference altitude, K is a correction factor, $H_{\text{sub.e}}$ is a scale height of the earth used to calculate a density of atmospheric air, $p_{\text{sub.0}}$ is an air density at an altitude of 0 m, $R_{\text{sub.e}}$ is an earth radius, $\beta_{\text{sub.m}}$ is a ballistic coefficient, $V_{\text{sub.e}}$ is the first cosmic velocity, and $L/D_{\text{sub.max}}$ is a maximum lift-drag ratio of the glide vehicle.
2. The glide control device according to claim 1, wherein the reference altitude is determined such that a gravity exerted on the glide vehicle by the earth, a lift force, and a centrifugal force are balanced in a vertical direction.
3. The glide control device according to claim 1, wherein an altitude upper limit value is defined for the reference altitude.
4. The glide control device according to claim 1, wherein the first cosmic velocity is calculated in accordance with the following expression (8):
$$V_{\text{sub.e}} = \sqrt{gR_{\text{sub.e}}}. \quad (8)$$
 where V_e is the first cosmic velocity, g is the acceleration of gravity, and R_e is an Earth radius.
5. A glide vehicle, comprising: an attitude control device; and a glide control device, the glide control device comprising: a communications device configured to communicate with the attitude control device of the glide vehicle in which no propulsion device configured to accelerate the glide vehicle is disposed; and a processor configured to control the attitude control device, wherein the processor is configured to control the attitude control device to generate a downward lift force so as to allow the glide vehicle to fly in a horizontal direction when a velocity of the glide vehicle is higher than or equal to a first cosmic velocity, and wherein the first cosmic velocity is a velocity

required by a spacecraft in a circular low Earth orbit, and wherein the processor is configured to: determine, based on the velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide; and control the attitude control device based on a current altitude of the glide vehicle and the reference altitude, wherein the reference altitude is calculated in accordance with the following expression (1): $h = KH_e \ln\left(\frac{\rho_0 R_e}{\beta_m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2} \cdot \text{Math.} \right)$, (1) wherein h is the reference altitude, K is a correction factor, H_e is a scale height of the earth used to calculate a density of atmospheric air, ρ_0 is an air density at an altitude of 0 m, R_e is an earth radius, β_m is a ballistic coefficient, V_e is the first cosmic velocity, and $L/D_{\max}$ is a maximum lift-drag ratio of the glide vehicle.

6. The glide vehicle according to claim 5, wherein the first cosmic velocity is calculated in accordance with the following expression (8):

$V_e = \sqrt{g R_e}$. (8) where V_e is the first cosmic velocity, g is the acceleration of gravity, and R_e is an Earth radius.

7. A flying body, comprising: an attitude control device; and a glide control device comprising: a communications device configured to communicate with the attitude control device of a glide vehicle in which no propulsion device configured to accelerate the glide vehicle is disposed; and a processor configured to control the attitude control device to generate a downward lift force so as to allow the glide vehicle to fly in a horizontal direction when a velocity of the glide vehicle is higher than or equal to a first cosmic velocity, which is a velocity required by a spacecraft in a circular low Earth orbit, a propulsion device configured to accelerate the glide vehicle; and a flight control device configured to: control the propulsion device to achieve a level flight at a release altitude; and release the glide vehicle at the release altitude, and wherein the processor is configured to: determine, based on the velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide; and control the attitude control device based on a current altitude of the glide vehicle and the reference altitude, wherein the reference altitude is calculated in accordance with the following expression (1):

$h = KH_e \ln\left(\frac{\rho_0 R_e}{\beta_m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2} \cdot \text{Math.} \right)$, (1) wherein h is the reference altitude, K is a correction factor, H_e is a scale height of the earth used to calculate a density of atmospheric air, ρ_0 is an air density at an altitude of 0 m, R_e is an earth radius, β_m is a ballistic coefficient, V_e is the first cosmic velocity, and $L/D_{\max}$ is a maximum lift-drag ratio of the glide vehicle.

8. The flying body according to claim 7, wherein the flight control device is configured to control the propulsion device such that a flying direction of the flying body is tilted downward with respect to a horizontal direction at the release altitude when the release altitude is an altitude out of the atmosphere.

9. The flying body according to claim 7, wherein the release altitude is an altitude in the atmosphere at which aerodynamic steering is achievable.

10. The flying body according to claim 7, wherein the release altitude comprises a reference altitude at which the glide vehicle should glide.

11. The flying body according to claim 7, wherein the flight control device is configured to control the propulsion device such that the flying body has a velocity higher than or equal to the first cosmic velocity at the release altitude.

12. The flying body accordingly to claim 7, wherein the first cosmic velocity is calculated in accordance with the following expression (8):

$V_e = \sqrt{g R_e}$. (8) where V_e is the first cosmic velocity, g is the acceleration of gravity, and R_e is an Earth radius.

13. A non-transitory tangible storage medium storing a program which when executed causes a processor to implement steps of: calculating, based on a velocity of a glide vehicle in which no

propulsion device configured to accelerate the glide vehicle is disposed, a reference altitude at which the glide vehicle should glide; and generating a glide control signal that controls an attitude of the glide vehicle based on the reference altitude, wherein generating the glide control signal comprises: generating the glide control signal to produce a downward lift force so as to allow the glide vehicle to fly in a horizontal direction when the velocity is higher than or equal to a first cosmic velocity, which is a velocity required by a spacecraft in a circular low Earth orbit, determining, based on the velocity of the glide vehicle, a reference altitude at which the glide vehicle should glide; and controlling the attitude based on a current altitude of the glide vehicle and the reference altitude, wherein the reference altitude is calculated in accordance with the following expression (1):
$$h = KH_e \ln\left(\frac{{}_0R_e}{m} L / D_{\max} \cdot \text{Math.} \frac{V^2}{V_e^2 - V^2} \cdot \text{Math.} \right), \quad (1)$$
 wherein h is the reference altitude, K is a correction factor, H.sub.e is a scale height of the earth used to calculate a density of atmospheric air, p.sub.0 is an air density at an altitude of 0 m, R.sub.e is an earth radius, β.sub.m is a ballistic coefficient, V.sub.e is the first cosmic velocity, and L/D.sub.max is a maximum lift-drag ratio of the glide vehicle.

14. The non-transitory tangible storage medium according to claim 13, wherein the first cosmic velocity is calculated in accordance with the following expression (8):

$$V_{\text{sub.e}} = \sqrt{gR_{\text{sub.e}}}. \quad (8)$$
 where V_e is the first cosmic velocity, g is the acceleration of gravity, and R_e is an Earth radius.
